

实验2.2

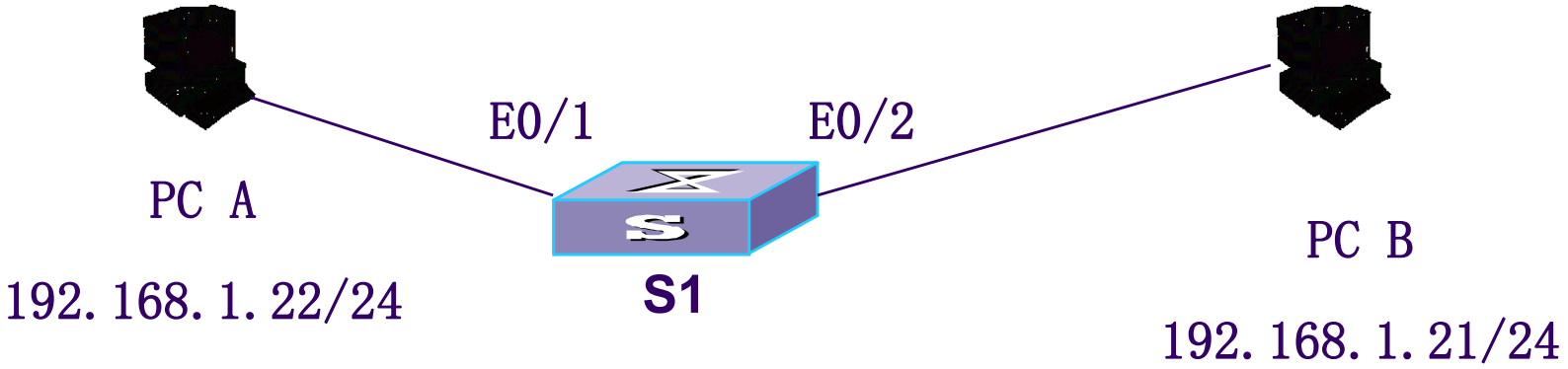
以太网帧格式分析

以太网链路层帧格式分析

- 实验目的
  - 分析比较Ethernet Ⅱ标准和IEEE 802.3标准的MAC帧格式

以太网链路层帧格式分析

◆实验组网图



PCA、PCB、S1的组网

以太网链路层帧格式分析

- 关键实验步骤说明
  - 按照组网图组网并配置各设备；
  - 在PCA 和PCB上启动Wireshark软件；
  - 截获Ethernet Ⅱ标准的MAC帧：

1.在PCA ping PCB；

2.停止截获报文，保存并分析；

Ethernet II 标准数据帧

No.	Time	Source	Destination	Protocol	Info
1	0.000000	HuaweiTe_5e:d4:b3	Broadcast	0x9001	Ethernet II
2	6.834021	50:78:1c:19:03:99	Broadcast	ARP	Who has 192.168.1.21? Tell 192.168.1.22
3	6.834200	50:78:1c:19:0d:04	50:78:1c:19:03:99	ARP	192.168.1.21 is at 50:78:1c:19:0d:04
4	6.834219	192.168.1.22	192.168.1.21	ICMP	Echo (ping) request
5	6.834345	192.168.1.21	192.168.1.22	ICMP	Echo (ping) reply
6	7.834238	192.168.1.22	192.168.1.21	ICMP	Echo (ping) request
7	7.834447	192.168.1.21	192.168.1.22	ICMP	Echo (ping) reply
8	8.834290	192.168.1.22	192.168.1.21	ICMP	Echo (ping) request
9	8.834486	192.168.1.21	192.168.1.22	ICMP	Echo (ping) reply
10	9.835165	192.168.1.22	192.168.1.21	ICMP	Echo (ping) request
11	9.835356	192.168.1.21	192.168.1.22	ICMP	Echo (ping) reply

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⊞ Frame 4 (74 bytes on wire, 74 bytes captured)
⊟ Ethernet II, Src: 50:78:1c:19:03:99 (50:78:1c:19:03:99), Dst: 50:78:1c:19:0d:04 (50:78:1c:19:0d:04)
  ⊞ Destination: 50:78:1c:19:0d:04 (50:78:1c:19:0d:04)
  ⊞ Source: 50:78:1c:19:03:99 (50:78:1c:19:03:99)
    Type: IP (0x0800)
⊞ Internet Protocol, Src: 192.168.1.22 (192.168.1.22), Dst: 192.168.1.21 (192.168.1.21)
⊞ Internet Control Message Protocol
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以太网链路层帧格式分析

- 关键实验步骤说明
  - 截获IEEE 802.3标准的MAC帧；

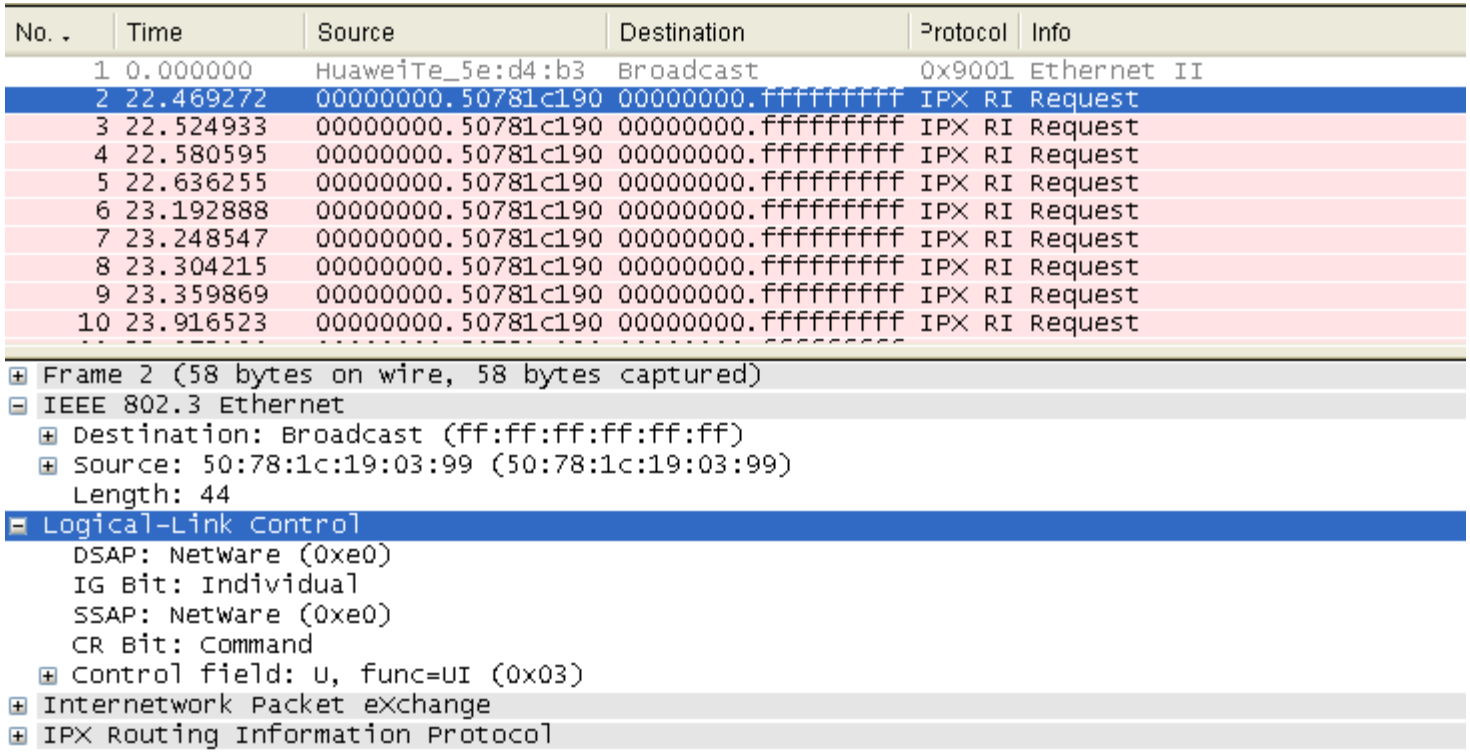
1. 打开Wireshark软件，并开始截获

2.在“本地连接属性”中配置添加协议：

NWLINK IPX/SPX/NETBIOS….

3.停止截获报文，保存并分析

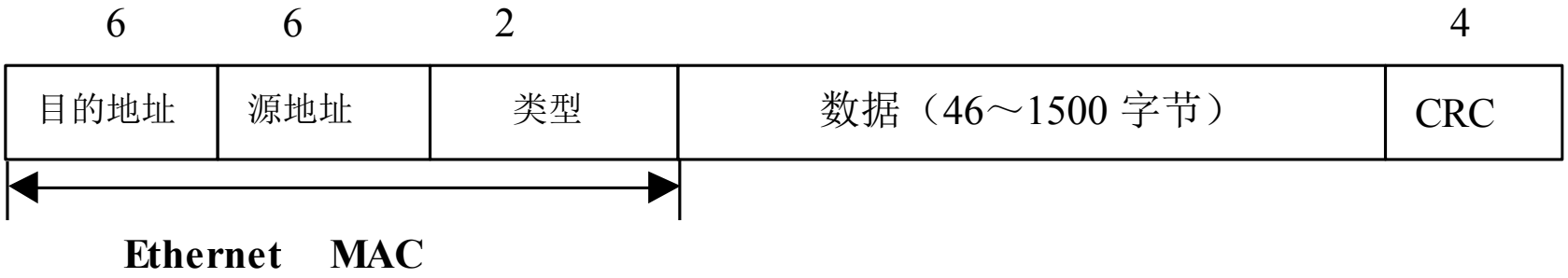
IEEE 802.3标准数据帧



No.	Time	Source	Destination	Protocol	Info
1	0.000000	HuaweiTe_5e:d4:b3	Broadcast	0x9001	Ethernet II
2	22.469272	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
3	22.524933	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
4	22.580595	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
5	22.636255	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
6	23.192888	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
7	23.248547	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
8	23.304215	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
9	23.359869	00000000.50781c190	00000000.ffffffffff	IPX RI	Request
10	23.916523	00000000.50781c190	00000000.ffffffffff	IPX RI	Request

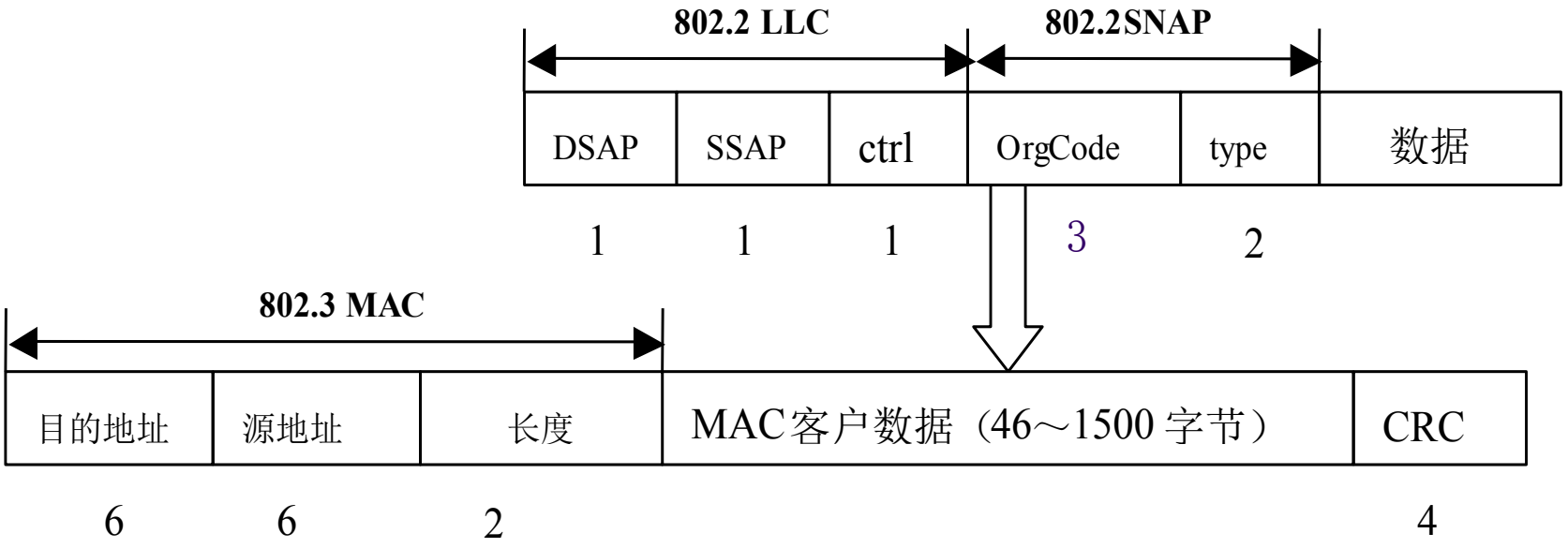
以太网链路层帧格式

- IEEE 802.3与EthernetⅡ封装数据的区别



Ethernet Ⅱ报文封装结构

以太网链路层帧格式



IEEE 802.3报文封装结构